

15.0 BUILT-IN SELF-TEST CONTROLLER

The ENC28J60 features a Built-in Self-Test (BIST) module which is designed to confirm proper operation of each bit in the 8-Kbyte memory buffer. Although it is primarily useful for testing during manufacturing, it remains present and available for diagnostic purposes by the user. The controller writes to all locations in the buffer memory and requires several pieces of hardware shared by normal Ethernet operations. Thus, the BIST should only be used on Reset or after necessary hardware is freed. When the BIST is used, the ECON1 register's DMAST, RXEN and TXRTS bits should all be clear.

The BIST controller is operated through four registers:

- EBSTCON register (control and status register)
- EBSTSD register (fill seed/initial shift value)
- EBSTCSH and EBSTCSL registers (high and low bytes of generated checksum)

The EBSTCON register (Register 15-1) controls the module's overall operation, selecting the Testing modes and starting the self-test process. The bit pattern for memory tests is provided by the EBSTSD seed register; its content is either used directly, or as the seed for a pseudo-random number generator, depending on the Test mode.

REGISTER 15-1: EBSTCON: ETHERNET SELF-TEST CONTROL REGISTER

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
PSV2	PSV1	PSV0	PSEL	TMSEL1	TMSEL0	TME	BISTST
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-5 **PSV2:PSV0:** Pattern Shift Value bits

When TMSEL<1:0> = 10:

The bits in EBSTSD will shift left by this amount after writing to each memory location.

When TMSEL<1:0> = 00, 01 or 11:

This value is ignored.

bit 4 **PSEL:** Port Select bit

1 = DMA and BIST modules will swap ports when accessing the memory

0 = Normal configuration

bit 3-2 **TMSEL1:TMSEL0:** Test Mode Select bits

11 = Reserved

10 = Pattern shift fill

01 = Address fill

00 = Random data fill

bit 1 **TME:** Test Mode Enable bit

1 = Enable Test mode

0 = Disable Test mode

bit 0 **BISTST:** Built-in Self-Test Start/Busy bit

1 = Test in progress; cleared automatically when test is done

0 = No test running

15.1 Using the BIST

When the BIST controller is started, it will fill the entire buffer with the data generated for the current test configuration and it will also calculate a checksum of the data as it is written. When the BIST is complete, the EBSTCS registers will be updated with the checksum. The host controller will be able to determine if the test passed or failed by using the DMA module to calculate a checksum of all memory. The resulting checksum generated by the DMA should match the BIST checksum. If after any properly executed test, the checksums differ, a hardware Fault may be suspected.

The BIST controller supports 3 different operations:

- Random Data Fill
- Address Fill
- Pattern Shift Fill

The ports through which the BIST and DMA modules access the dual port SRAM can be swapped for each of the four Test modes to ensure proper read/write capability from both ports.

To use the BIST:

1. Program the EDMAST register pair to 0000h.
2. Program EDMAND and ERXND register pairs to 1FFFh.
3. Configure the DMA for checksum generation by setting CSUMEN in ECON1.
4. Write the seed/initial shift value byte to the EBSTSD register (this is not necessary if Address Fill mode is used).
5. Enable Test mode, select the desired test, select the desired port configuration for the test.
6. Start the BIST by setting EBSTCON.BISTST.
7. Start the DMA checksum by setting DMAST in ECON1. The DMA controller will read the memory at the same rate the BIST controller will write to it, so the DMA can be started any time after the BIST is started.
8. Wait for the DMA to complete by polling the DMAST bit or receiving the DMA interrupt (if enabled).
9. Compare the EDMACS registers with the EBSTCS registers.

To ensure full testing, the test should be redone with the Port Select bit, PSEL, altered. When not using Address Fill mode, additional tests may be done with different seed values to gain greater confidence that the memory is working as expected.

At any time during a test, the test can be canceled by clearing the BISTST, DMAST and TME bits. While the BIST is filling memory, the EBSTSD register should not be accessed, nor should any configuration changes occur. When the BIST completes its memory fill and checksum generation, the BISTST bit will automatically be cleared.

The BIST module requires one main clock cycle for each byte that it writes into the RAM. The DMA module's checksum implementation requires the same time but it can be started immediately after the BIST is started. As a result, the minimum time required to do one test pass is slightly greater than 327.68 μ s.

15.2 Random Data Fill Mode

In Random Data Fill mode, the BIST controller will write pseudo-random data into the buffer. The random data is generated by a Linear Feedback Shift Register (LFSR) implementation. The random number generator is seeded by the initial contents of the EBSTSD register and the register will have new contents when the BIST is finished.

Because of the LFSR implementation, an initial seed of zero will generate a continuous pattern of zeros. As a result, a non-zero seed value will likely perform a more extensive memory test. Selecting the same seed for two separate trials will allow a repeat of the same test.

15.3 Address Fill Mode

In Address Fill mode, the BIST controller will write the low byte of each memory address into the associated buffer location. As an example, after the BIST is operated, the location 0000h should have 00h in it, location 0001h should have 01h in it, location 0E2Ah should have 2Ah in it and so on. With this fixed memory pattern, the BIST and DMA modules should always generate a checksum of F807h. The host controller may use Address Fill mode to confirm that the BIST and DMA modules themselves are both operating as intended.

15.4 Pattern Shift Fill Mode

In Pattern Shift Fill mode, the BIST controller writes the value of EBSTSD into memory location 0000h. Before writing to location 0001h, it shifts the contents of EBSTSD to the left by the value specified by the PSV2:PSV0 bits in EBSTCON. Bits that leave the most significant end of EBSTSD are wrapped around to the least significant side. This shift is repeated for each new address. As a result of shifting the data, a checkerboard pattern can be written into the buffer memory to confirm that adjacent memory elements do not affect each other when accessed.

TABLE 15-1: SUMMARY OF REGISTERS ASSOCIATED WITH THE SELF-TEST CONTROLLER

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
ECON1	TXRST	RXRST	DMAST	CSUMEN	TXRTS	RXEN	BSEL1	BSEL0	13
ERXNDL	RX End Low Byte (ERXND<7:0>)								13
ERXNDH	—	—	—	RX End High Byte (ERXND<12:8>)					13
EDMASTL	DMA Start Low Byte (EDMAST<7:0>)								13
EDMASTH	—	—	—	DMA Start High Byte (EDMAST<12:8>)					13
EDMANDL	DMA End Low Byte (EDMAND<7:0>)								13
EDMANDH	—	—	—	DMA End High Byte (EDMAND<12:8>)					13
EDMACSL	DMA Checksum Low Byte (EDMACS<7:0>)								13
EDMACSH	DMA Checksum High Byte (EDMACS<15:8>)								13
EBSTSD	Built-in Self-Test Fill Seed (EBSTSD<7:0>)								14
EBSTCON	PSV2	PSV1	PSV0	PSEL	TMSEL1	TMSEL0	TME	BISTST	14
EBSTCSL	Built-in Self-Test Checksum Low Byte (EBSTCS<7:0>)								14
EBSTCSH	Built-in Self-Test Checksum High Byte (EBSTCS<15:8>)								14

Legend: — = unimplemented, read as '0'. Shaded cells are not used.

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NOTES: